DAI 701 · RESEARCH METHODOLOGIES IN AI · MIDTERM ASSIGNMENT

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Systematic Literature Review

A publishable-quality survey of your approved AI topic

ASSESSMENT	WEIGHT	DUE	FORMAT
Midterm	20 points	Week 8–9	IEEE / ACM / Springer

1 Overview

This is one of the main assignment of the course and serves as your midterm. You will produce a complete, publishable-quality systematic literature review (SLR) on the AI topic approved in Week 3 — for example, one of the four areas explored in class: AI in Education, Blockchain & AI, Generative AI in Business, or Machine Learning in Healthcare. The review must be reproducible, critically synthesised, and formatted to a real publisher template.

A systematic review is not a summary of papers you read. It is an argument built from evidence: it defines a question, searches for evidence in a documented and reproducible way, screens it against explicit criteria, organises it into a taxonomy, critically synthesises it, and derives defensible research gaps. A paper-by-paper enumeration — however many references it cites — will not pass the synthesis criterion.

2 What you will demonstrate

- The ability to execute and document a reproducible scholarly search (PRISMA-2020 + Kitchenham).
- Screening a large corpus to a defensible working set with explicit inclusion/exclusion and quality criteria.
- Synthesising a body of work into an original taxonomy and a critical, comparative analysis.
- Deriving non-trivial research gaps and articulating a forward-looking research agenda.
- Producing a correctly formatted, correctly cited manuscript in a standard publisher template.

3 Required structure

Your manuscript must contain the following sections, in this order. Section 5 (Analysis by category) is the core and carries the most weight.

Section	≈ Length	Status	What it contains
Title, abstract, keywords	150–250 w	Required	Precise scope; the survey's contributions; 4–6 keywords. The abstract states RQs, method, and headline findings.
1. Introduction	1.5–2 pp	Required	Motivation; scope and boundaries; the review's research questions; why existing surveys are insufficient; contributions; roadmap.
2. Background	1–2 pp	Required	Minimal preliminaries and terminology a reader needs to follow the survey. Not a tutorial — keep it lean.
3. Review methodology	1–1.5 pp	Required	The SLR protocol: databases and exact search strings; inclusion/exclusion and quality criteria; PRISMA-2020 flow diagram; how papers were coded; screening reliability.
4. Taxonomy / classification	3–5 pp	Required	The organizing framework; dimension definitions; a taxonomy figure; mapping of works to categories.
5. Analysis by category	6–12 pp	Required	THE CORE. Critical synthesis of each branch; comparison tables; trends; strengths and limitations. Not paper-by-paper summary.
6. Datasets, benchmarks & evaluation	1.5–3 pp	Required	What the field trains and tests on; metrics; reproducibility status.
7. Open problems & research agenda	2–3 pp	Required	Gaps derived from empty taxonomy cells and critique; concrete future directions; ethics/fairness/reproducibility challenges.
8. Conclusion	0.5 p	Required	What the survey establishes, and for whom. No new material.
References	—	Required	Complete, consistent, machine-checked (BibTeX). ≥ 80 references (target 100–150).

4 Optional sections you may add

Strong reviews often include one or more of the following. Add them where they strengthen the argument — not to inflate length. Choose deliberately.

Optional section	When to include it
Related / prior surveys	A dedicated section (or subsection of the Introduction) comparing your review with existing surveys in a table — what they covered, their gaps, and what yours adds.
Bibliometric & trend analysis	Publications per year, leading venues, author/institution or geographic distribution, citation trends. Strengthens claims about where the field is heading.
Historical evolution / timeline	A figure tracing the field's milestones when the story is strongly chronological.
Theoretical / conceptual framework	When the synthesis is organised around a theory or model rather than a purely technical taxonomy.
Applications & use cases	A section surveying real-world deployments when the field is application-driven.
Tools, frameworks & platforms	A landscape of the software/tooling ecosystem, often as a comparison table.
Discussion	A separate interpretive section distinct from the per-category analysis — cross-cutting themes, contradictions, and implications.
Threats to validity of the review	A short, honest account of your review's own limitations (search bias, single-screener bias, language/venue bias). Highly valued at PhD level.
Glossary / list of acronyms	When the field is acronym-heavy.
Appendices	Full search strings per database, the complete extraction table, quality-assessment scores, and the PRISMA-2020 checklist.
Reproducibility / data-availability statement	A pointer to the shared BibTeX, extraction sheet, and any scripts — expected in modern venues.
Front-matter (publication style)	Author contributions, funding, conflict-of-interest, and acknowledgements if you format it as a submission-ready manuscript.

5 Methodology requirements

Search (Week 4 work)

Start from concepts, not keywords: identify the core concepts of your question, generate synonyms, and build a search string. Run the same query across at least three databases (e.g. Scopus / Web of Science, Semantic Scholar, IEEE Xplore/ACM DL, OpenAlex, Google Scholar) and record exact strings, date ranges, and hit counts per database. Deduplicate in a reference manager and add backward/forward snowballing.

Screening & the PRISMA-2020 flow (Week 5 work)

Apply explicit inclusion/exclusion and quality criteria, and report the funnel as a PRISMA-2020 flow diagram with real numbers (identified → screened → eligible → included). State how disagreements were resolved and, where relevant, a screening-reliability measure.

Extraction & synthesis (Weeks 6–7 work)

Extract a consistent set of fields per paper (problem, RQ, dataset/sample, method, findings, limitations, gap). Turn the extraction table into a 2–3 dimension taxonomy — empty cells are candidate gaps. Then synthesise: integrate, compare, and explain, rather than listing.

6 Length, references & quality bar

Requirement	Specification
Length	18–30 pages in the chosen template (excluding references), or ≈ 8,000–12,000 words.
Papers screened	150–300 works screened; the corpus and PRISMA counts must be documented.

Requirement	Specification
References cited	≥ 80 (target 100–150), mixing seminal and current work; ≥ 40% from the last three years.
Original synthesis	≥ 1 non-trivial taxonomy figure and ≥ 2 comparison tables that are YOUR synthesis, not copied.
Critical depth	Every category ends with analysis (limitations, open questions) — never mere enumeration.
Reproducibility	The search is reproducible from the documented protocol; shared BibTeX + extraction sheet accompany the paper.
Integrity	No plagiarism; all AI-tool assistance disclosed; every claim traceable to a cited source.

7 Formatting & template

Choose ONE standard publisher template and apply it consistently throughout:

- **IEEE** — IEEEtran (conference/journal); the default for most AI venues.
- **ACM** — acmart (sigconf); common for computing surveys (e.g. ACM Computing Surveys).
- **Springer** — LNCS / svjour3; used across Springer journals and proceedings.

LaTeX is strongly preferred (Use Overleaf platform) and references must be managed with BibTeX. Word is permitted only by exception and must still match the chosen template's layout. Do not mix templates. Figures and tables must be captioned, numbered, and referenced in the text.

8 The three exemplar reviews

Three published systematic reviews are provided with this assignment. Study them as models of structure, taxonomy design, and synthesis depth — not as text to imitate or paraphrase. From each, extract: (a) how they scope and justify the review, (b) how they present the PRISMA flow, (c) how their taxonomy figure is built, and (d) how a category section moves from evidence to critique.

#	Exemplar (instructor to provide title/venue)	What to study in it
1	From Language to Action: A Review of Large Language Models as Autonomous Agents and Tool Users	Overall structure & scoping of the review.
2	Beyond one-shot explanations: a systematic literature review of dialogue-based xAI approaches	Taxonomy figure and category-by-category synthesis.
3	A systematic review of multi-modal large language models on domain-specific applications	PRISMA method reporting and the research-agenda section.

USE THE EXEMPLARS AS SCAFFOLDING, NOT AS SOURCE TEXT

Reproducing the exemplars' wording, structure, or figures is plagiarism. Learn the *moves* they make — then make those moves on your own corpus, in your own words.

9 Grading — 20 points

The synthesis and the research gap carry the most weight, reflecting what distinguishes a review from a summary. A paper-by-paper survey cannot score highly on synthesis even if it cites many sources.

Criterion	Pts	What full marks look like
Search & reproducibility	**2**	Exact strings, databases, dates, and per-database counts; a correct PRISMA-2020 flow with real numbers; screening reliability addressed.
Coverage & source quality	**2**	80+ apt references spanning seminal and recent (≥40% last 3 yrs); appropriate venues; no padding.
Taxonomy / classification	**3**	A clear, justified organizing framework with defined dimensions and an original taxonomy figure mapping works to categories.
Critical synthesis (core)	**5**	Evidence is integrated, compared, and interpreted — agreements, contradictions, and why results diverge. Not a list of paper summaries.
Research gap & agenda	**3**	Defensible, non-trivial gaps derived from the synthesis; a concrete, forward-looking research agenda.
Methodological rigor	**2**	Protocol elements complete and internally consistent; inclusion/exclusion and quality criteria justified; threats to validity acknowledged.
Writing, structure & template	**2**	Clear scholarly prose; correct IEEE/ACM/Springer formatting; figures and tables captioned and referenced.
Citations & integrity	**1**	Consistent citation style; complete BibTeX; AI-use disclosure present; all claims traceable.
TOTAL	**20**	Midterm weight of the course.

10 Suggested topics

	Suggested topic	SLR suggested Title
1	Agentic AI for Autonomous Cybersecurity: From Intrusion Detection to Autonomous Incident Response	Agentic AI for Autonomous Cybersecurity Operations: A Systematic Review of LLM-Based Intrusion Detection, Threat Hunting, and Incident Response
2	Agentic AI for Dynamic Scheduling and Resource Allocation in IoT/Edge Systems	Agentic AI for Dynamic Scheduling and Resource Allocation in IoT and Edge Computing: A Systematic Review and Research Agenda
3	Agentic AI for Healthcare Decision-Making: Reliability, Explainability and Autonomous Clinical Workflows	Agentic AI in Healthcare Decision-Making: A Systematic Review of Autonomous Clinical Agents, Reliability, Explainability, and Human Oversight
4	Agentic AI for Financial Decision-Making	Agentic Artificial Intelligence in Financial Decision-Making: A Systematic Literature Review of Architectures, Applications, and Challenges.
5	Multi-Agent AI for Supply Chain Optimization	Multi-Agent Artificial Intelligence for Supply Chain Optimization: A Systematic Literature Review of Architectures, Coordination Mechanisms, and Applications.
6	Agentic AI for Time-Series Forecasting and Anomaly Detection	Agentic Artificial Intelligence for Time-Series Forecasting and Anomaly Detection: A Systematic Literature Review of Architectures, Methods, and Applications.
7	Agentic AI for Business Performance Management	Agentic Artificial Intelligence for Business Performance Management: A Systematic Literature Review of Autonomous Decision-Making, Performance Monitoring, and Optimization.
8	Generative AI for Business Intelligence	Generative Artificial Intelligence in Business Intelligence: A Systematic Literature Review of Applications, Capabilities, and Emerging Challenges.
9	Agentic AI for Business Intelligence and Decision Support	Agentic AI for Business Intelligence and Decision Support: A Systematic Literature Review of Autonomous Analytics, Multi-Agent Architectures, and Enterprise Applications.
10	From Traditional BI to Agentic Business Intelligence	From Business Intelligence to Agentic Business Intelligence: A Systematic Literature Review of Autonomous Analytics, Decision Support, and Business Performance Management.

11 What to submit

- The manuscript as a PDF built from the IEEE / ACM / Springer template. Overleaf project must be shared with me.
- The BibTeX (.bib) file for all references.
- The extraction spreadsheet (one row per included paper).
- The PRISMA-2020 flow diagram and the completed PRISMA checklist (may be an appendix).
- An AI-use disclosure statement describing any tool assistance.
- **All above must be submitted via Moodle platform.**

12 How the course builds to this

Each weekly milestone is a component of the final review. If you keep pace, this assignment is assembled from work you have already done and graded.

Week	Milestone feeding this assignment
Wk 3	Survey topic selected and approved (gate).
Wk 4	Reproducible search strategy drafted; corpus imported and deduplicated (BibTeX).
Wk 5	SLR protocol + PRISMA-2020 diagram submitted.
Wk 6	Annotated bibliography; taxonomy and candidate-gap list drafted.
Wk 7	Review RQs finalized; Introduction drafted.
Wk 8–9	FULL SYSTEMATIC REVIEW DUE — this assignment (midterm).

ACADEMIC INTEGRITY & RESPONSIBLE AI USE

Every claim must be traceable to a cited source. Do not fabricate references or paraphrase the exemplars. AI tools may assist with editing, searching, or drafting, but their use must be disclosed, and you are fully responsible for the accuracy of every citation and statement. Undisclosed AI-generated content or plagiarism results in a failing grade for the midterm.

13 Submission checklist

- ☐ Question, scope, and RQs are explicit and answerable.
- ☐ Search is reproducible: exact strings, databases, dates, and counts documented.
- ☐ PRISMA-2020 flow diagram with real numbers included.
- ☐ ≥ 80 references (target 100–150), ≥ 40% from the last three years.
- ☐ An original taxonomy figure and ≥ 2 original comparison tables.
- ☐ Every category section ends in critique, not enumeration.
- ☐ Defensible research gaps and a concrete research agenda.
- ☐ Formatted in one publisher template; figures/tables captioned and referenced.
- ☐ BibTeX, extraction sheet, and AI-use disclosure attached.